

STUDENT DOCUMENT SIMILARITY ANALYSIS

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Data Structures and Algorithms -3 (25CS2103E)

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DECLARATION

The Mini Project Report entitled "Student Document Similarity Analysis" is a record of bonafide work of J. Simran - 2520030166, K. Anusri - 2520030432, and S. Rikhil - 2520030480, submitted in partial fulfillment for the award of B.Tech in Computer Science and Engineering to the K L (Deemed to be) University. The results embodied in this report have not been copied from any other department, university, or institute.

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CERTIFICATE

This is to certify that the mini project based report entitled "Student Document Similarity Analysis" is a bonafide work done and submitted by J. Simran - 2520030166, K. Anusri - 2520030432, and S. Rikhil - 2520030480 in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in Computer Science Engineering, K L (Deemed to be University), during the academic year 2026-2027.

Signature of the Supervisor

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Signature of the External Examiner

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1.ABSTRACT

The project is a Student Document Similarity Analysis system designed to automatically identify similarities between multiple student documents. Comparing a large number of documents manually is time-consuming and may result in difficulty identifying copied or repeated content. The proposed system processes a collection of documents from a corpus, converts the text into lowercase, and divides the content into meaningful sentence chunks. Every possible pair of documents is then compared using the Z-Algorithm for exact substring matching.

The Z-Algorithm constructs a Z-array that stores the length of the longest substring starting from each position that matches the prefix of the combined pattern and text. The system uses this algorithm to identify matching content efficiently and calculates a similarity percentage for each document pair. The similarity results are classified according to predefined thresholds and arranged from highest to lowest similarity.

The system displays the document pairs having significant similarity along with their similarity percentage and verdict. Documents below the minimum similarity threshold are not displayed in the similar-file results. The project provides an efficient approach for reducing the effort required for manual document comparison and can be useful for identifying copied or repeated content in student submissions.

This project demonstrates the practical application of the Z-Algorithm in document similarity analysis and shows how an efficient string-matching algorithm can be applied to a real-world problem.

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2. INTRODUCTION

A large number of student documents can be difficult to compare manually, especially when dealing with a large collection. Students may submit documents containing copied or repeated content, making it necessary to identify similar documents efficiently.

Checking every document against all other documents manually requires considerable time and effort. Our system automatically processes a collection of documents from a corpus, converts the text into lowercase, and divides it into meaningful sentence chunks. Every possible pair of documents is compared using the Z-Algorithm, and a similarity percentage is calculated for each relevant pair. The results are then sorted from highest similarity to lowest similarity, making it easier to identify documents containing significant common or duplicated content.

The main focus of the project is exact text matching using the Z-Algorithm. The approach provides an efficient way to locate matching substrings and compare document content without requiring advanced machine-learning models.

The complete system consists of document reading, text preprocessing, sentence chunking, pairwise comparison, similarity calculation, threshold filtering, and result sorting.

The literature survey focuses on existing techniques used for plagiarism and document similarity detection. Recent systems use natural language processing, machine learning, transformer models and semantic analysis. The following studies provide background for understanding the existing approaches and the research gap addressed by this project.

1. Plagiarism Detection in the Digital Era (2024): This work reviewed modern plagiarism detection methods and emphasized the need for automated, reliable and efficient document comparison systems.
2. T-SRE: Transformer-based Semantic Relation Extraction (2024): This approach used transformer-based analysis to detect paraphrased plagiarism and semantic similarities that may not be detected by traditional exact text matching techniques.
3. Plagiarism Detection Using Machine Learning (2024): This study explored natural language processing and machine-learning techniques for detecting exact and paraphrased plagiarism with improved accuracy and scalability.
4. Comparative Analysis of Text-Based Plagiarism Detection Techniques (2025): This study compared different text-based plagiarism detection methods and discussed techniques for improving similarity detection.

Research Gap / Our Approach: Recent methods focus on NLP, machine learning and semantic analysis. Our project uses the Z-Algorithm for exact text matching, compares multiple documents pairwise, calculates similarity percentages and ranks the results to identify highly similar document pairs.

3. LITERATURE SURVEY

3.1 Plagiarism Detection in the Digital Era

Modern plagiarism detection systems have been developed to automatically identify copied content from digital documents. These systems emphasize automated and efficient comparison methods because manual comparison becomes difficult when the number of documents increases. This motivates the development of automated document similarity systems such as the proposed project.

3.2 T-SRE: Transformer-based Semantic Relation Extraction

Transformer-based approaches can analyze semantic relationships between text segments. Such methods can detect paraphrased plagiarism and semantic similarities that traditional exact text-matching techniques may fail to identify. However, these approaches are more complex than direct string matching.

3.3 Plagiarism Detection Using Machine Learning

Machine-learning and Natural Language Processing techniques have been explored for detecting exact and paraphrased plagiarism. These approaches can provide improved accuracy and scalability, particularly when documents contain modified versions of the same information.

3.4 Comparative Analysis of Text-Based Plagiarism Detection Techniques

Different text-based plagiarism detection techniques have been compared to understand their effectiveness in identifying similar content. Such studies demonstrate the importance of selecting appropriate similarity detection methods based on the type of documents and matching required.

3.5 Research Gap

Recent approaches increasingly use Natural Language Processing, machine learning, and semantic analysis. The proposed project instead focuses on the **Z-Algorithm for exact text matching**. It compares multiple documents, calculates similarity percentages, and ranks document pairs based on their similarity.

The main focus is therefore on demonstrating an efficient and understandable Data Structures and Algorithms approach to document similarity analysis.

4. Problem Statement and Objectives

4.1 Problem Statement

When a large number of student documents are submitted, manually comparing every document with the others is time-consuming and inefficient. Some documents may contain copied or repeated content with only minor changes. Therefore, an automated system is required to compare multiple documents, identify similar document pairs, and calculate their similarity percentage.

4.2 Problem Description

The system automatically processes documents from a corpus and compares every possible document pair using exact string matching. It identifies common content, calculates a similarity percentage, and displays document pairs that satisfy the minimum similarity threshold.

4.3 Objectives

The main objectives of the project are:

1. To process multiple student documents from a document corpus.
2. To preprocess the document text before comparison.
3. To convert document content into lowercase.
4. To divide documents into meaningful sentence chunks.
5. To compare every possible pair of documents.
6. To implement the Z-Algorithm for exact substring matching.
7. To calculate the similarity percentage between document pairs.
8. To classify documents based on their similarity.
9. To filter out document pairs below the minimum similarity threshold.
10. To display similar document pairs in descending order of similarity.
11. To reduce the manual effort required for comparing large numbers of documents.

5. HARDWARE AND SOFTWARE REQUIREMENTS

5.1 Hardware Requirements

Hardware	Requirement
Computer/Laptop	Required
Processor	Modern processor
RAM	Minimum 4 GB
Storage	Sufficient storage for document corpus
Keyboard	Required
Mouse	Required

5.2 Software Requirements

Software	Requirement
Operating System	Windows / Linux
Programming Language	Java
File Handling	Java I/O and NIO
IDE / Code Editor	Any Java-compatible IDE
Algorithm	Z-Algorithm

6. IMPLEMENTATION

6.1 Module 1: Document Input and File Handling

The first module is responsible for reading documents from the corpus folder.

The system scans the corpus directory and identifies the available .txt files. Each document is read using Java file-handling facilities. The content of every document is stored for further processing.

Main operations

- Access corpus directory.
- Identify .txt files.
- Read document contents.
- Store document data for processing.

The project uses Java I/O and NIO file handling for this purpose.

6.2 Module 2: Text Preprocessing

Before comparing documents, the text is pre-processed.

The system converts document content into lowercase so that differences in uppercase and lowercase letters do not affect exact matching.

The documents are then divided into meaningful sentence chunks. These chunks are used during the similarity comparison process.

Preprocessing steps

Input Document



Read Text



Convert to Lowercase



Divide into Sentence Chunks



Store Processed Chunks

This preprocessing step makes the document comparison more consistent.

6.3 Module 3: Z-Algorithm Based Matching

The core algorithm of the project is the **Z-Algorithm**.

The Z-Algorithm is a string-matching algorithm that identifies occurrences of a pattern inside a text efficiently.

For a string:

$S = \text{Pattern} + \text{Special Character} + \text{Text}$

the algorithm constructs a **Z-array**.

The Z-value at a position represents the length of the longest substring starting from that position that matches the prefix of the combined string.

When the Z-value becomes equal to the length of the pattern, a complete pattern match has been found.

Example

Suppose:

Pattern = "data"

Text = "This data structure uses data"

The algorithm searches for occurrences of the pattern in the text and identifies matching positions.

The Z-Algorithm works in:

$O(n + m)$

where:

- n = length of the text
- m = length of the pattern

The original abstract describes this linear-time property and the construction of the Z-array.

6.4 Module 4: Document Pair Comparison

After preprocessing, the system compares every possible pair of documents.

For example, if the corpus contains:

Document 1

Document 2

Document 3

Document 4

the system compares:

Document 1 ↔ Document 2

Document 1 ↔ Document 3

Document 1 ↔ Document 4

Document 2 ↔ Document 3

Document 2 ↔ Document 4

Document 3 ↔ Document 4

For each pair, the system identifies matching chunks using the Z-Algorithm and calculates a similarity percentage.

6.5 Module 5: Similarity Calculation

After matching the document chunks, the system calculates a similarity percentage.

The similarity value represents the amount of common content identified between the compared documents.

The calculated similarity percentage is then used to classify the document pair.

Similarity Classification

Similarity	Verdict
------------	---------

80% and above	High Chance of Copying - Nearly Identical
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50% - 79.99%	Suspiciously Similar - Heavy Overlap
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30% - 49.99%	Related - Some Shared Content
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Below 30%	Not displayed
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6.6 Module 6: Result Sorting and Display

After calculating the similarity percentages, the document pairs are sorted from highest similarity to lowest similarity.

The system displays:

- Pair of documents
- Similarity percentage
- Similarity verdict

Only document pairs meeting the minimum similarity threshold are displayed in the similar-file results.

Z-Algorithm:

Input

- Pattern P
- Text T

Process

1. Combine pattern and text using a special delimiter.
2. Construct the combined string.
3. Create the Z-array.
4. Maintain the Z-box using left and right boundaries.
5. Compare characters inside and outside the current Z-box.
6. Store the length of the matching prefix at every position.
7. Identify positions where the Z-value equals the pattern length.
8. Mark those positions as successful matches.

Output

- Matching positions
- Matching substrings
- Similarity information

Time Complexity

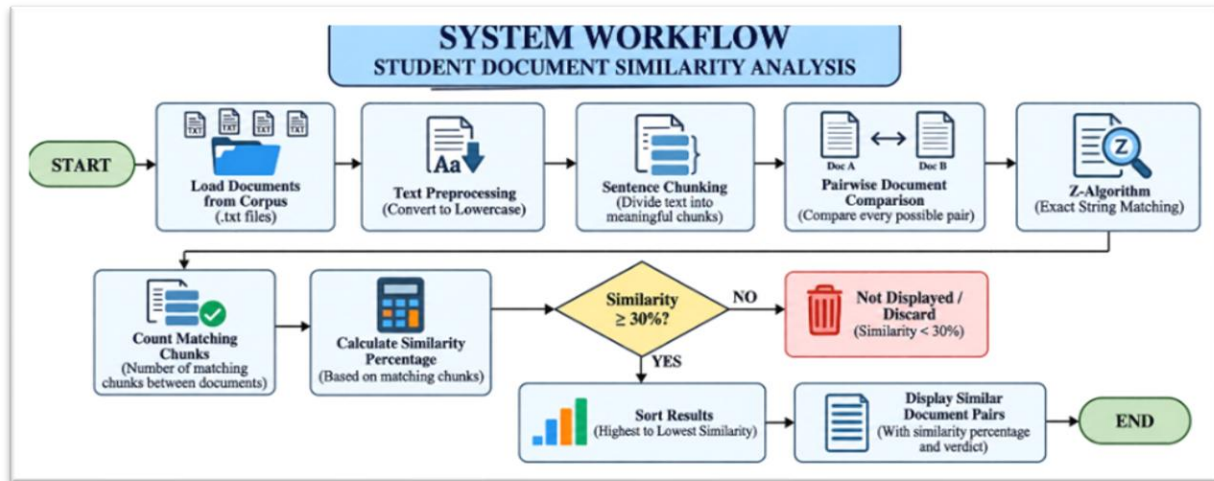
$O(n + m)$

Space Complexity

$O(n + m)$

where n and m represent the lengths of the text and pattern.

System Workflow:



7. RESULTS & DISCUSSION

The developed system successfully processes a collection of documents and compares every possible document pair.

For each similar document pair, the system displays:

1. **Document Pair**
2. **Similarity Percentage**
3. **Similarity Verdict**

The results are arranged from highest similarity to lowest similarity. This allows users to identify highly similar documents quickly without manually inspecting every document.

Similarity Results

The system uses the following classification:

80% and Above

HIGH CHANCE OF COPYING - Nearly Identical

Documents in this category contain a very high amount of common content.

50% - 79.99%

SUSPICIOUSLY SIMILAR - Heavy Overlap

Documents in this range contain substantial common content.

30% - 49.99%

RELATED - Some Shared Content

Documents have noticeable common content but are not considered highly similar.

Below 30%

These document pairs are not displayed in the similar-file results.

DISCUSSION

The project demonstrates that the Z-Algorithm can be effectively applied to exact text matching in document comparison.

The main advantage of the approach is that it uses an efficient string-matching algorithm rather than relying on manual comparison. The preprocessing stage provides a consistent representation of document text, while pairwise comparison allows the system to analyze multiple documents.

The similarity threshold prevents documents with very low similarity from appearing in the final similar-document results. Sorting the results makes it easier to prioritize documents with higher similarity.

However, the current approach primarily focuses on **exact text matching**. Therefore, documents that express the same information using significantly different wording or paraphrasing may not be identified as highly similar

8. CONCLUSION & FUTURE SCOPE

The **Student Document Similarity Analysis** project provides an automated approach for identifying similar content between multiple student documents.

The system preprocesses documents, compares every possible document pair, and uses the **Z-Algorithm** for exact substring matching. Similarity percentages are calculated for the document pairs, and the results are filtered using a minimum similarity threshold.

The system displays the relevant document pairs along with their similarity percentages and verdicts and sorts the results from highest to lowest similarity.

The project successfully demonstrates the practical application of the Z-Algorithm to a real-world document comparison problem. It reduces the manual effort required to compare large numbers of documents and provides a structured way to identify documents containing significant common or duplicated content.

FUTURE SCOPE

The project can be enhanced in several ways.

13.1 PDF and DOCX Support

The system can be extended to directly process PDF and DOCX files instead of only text files.

13.2 Large-Scale Document Comparison

The system can be optimized to compare a much larger number of documents efficiently.

13.3 Graphical User Interface

A user-friendly GUI can be developed to allow users to upload documents and view similarity results easily.

13.4 Match Highlighting

The system can highlight the exact portions of documents that match, making it easier to understand why two documents are considered similar.

13.5 Detailed Similarity Reports

The application can generate downloadable reports containing document pairs, similarity percentages, matched content, and verdicts.

13.6 Paraphrase Detection

Advanced Natural Language Processing techniques can be added to detect content that has been rewritten or paraphrased.

13.7 Document Storage

Previously uploaded documents and their comparison results can be stored for future analysis.

13.8 Advanced Similarity Techniques

Additional text similarity algorithms can be integrated to improve the system beyond exact

9. REFERENCES

1. **Plagiarism Detection in the Digital Era, 2024.**
Reference URL: *Insert the actual source URL used for the literature survey.*
2. **T-SRE: Transformer-based Semantic Relation Extraction, 2024.**
Reference URL: *Insert the actual source URL used.*
3. **Plagiarism Detection Using Machine Learning, 2024.**
Reference URL: *Insert the actual source URL used.*
4. **Comparative Analysis of Text-Based Plagiarism Detection Techniques, 2025.**
Reference URL: *Insert the actual source URL used.*
5. **Z-Algorithm / String Matching Algorithm reference.**
Reference URL: *Insert the actual source U*





